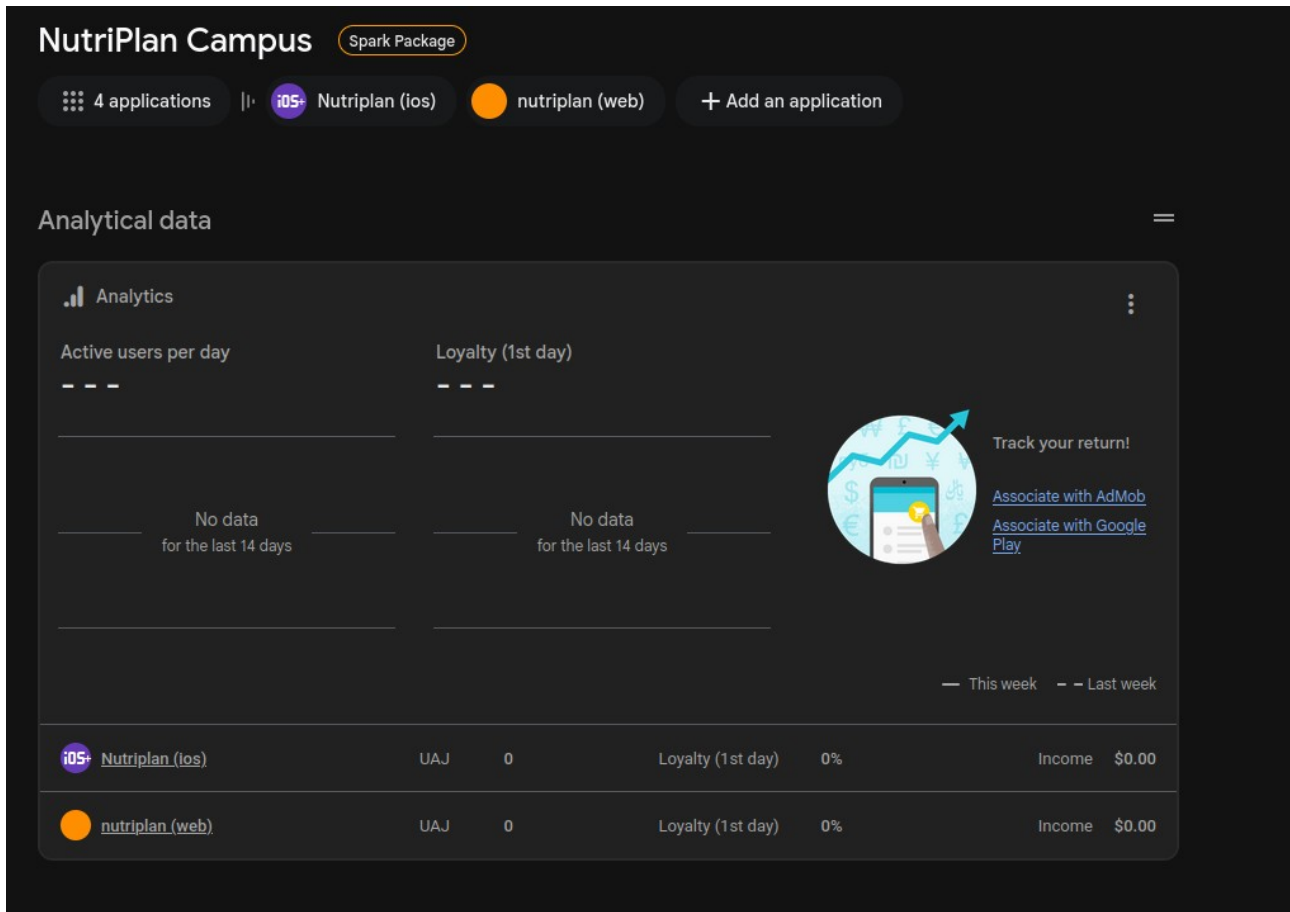


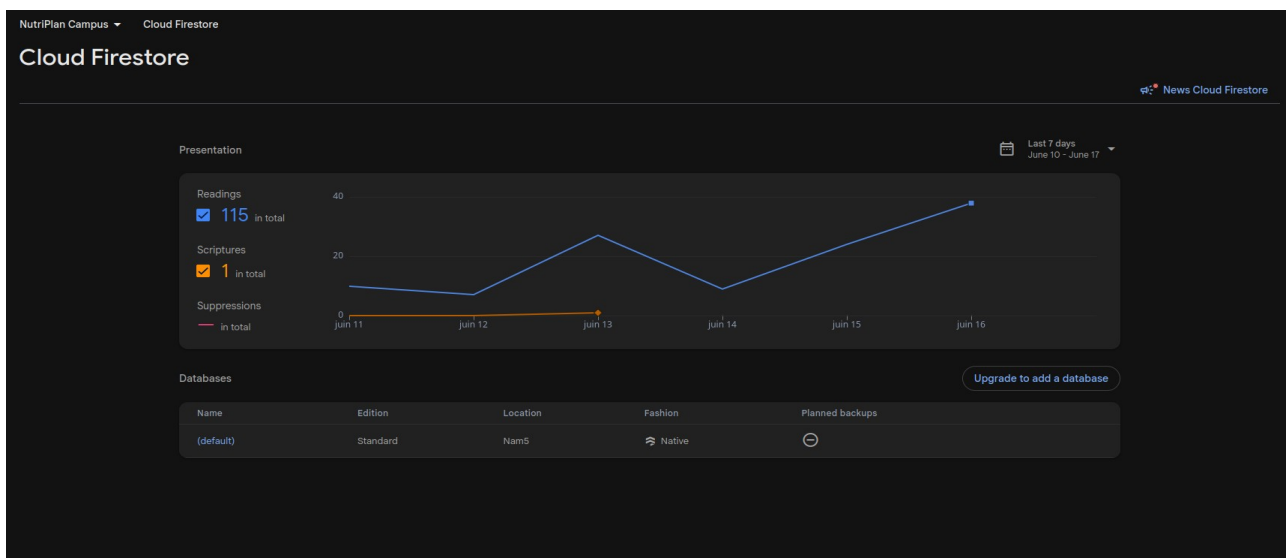
Week 6 and 7

Figure 1: Firebase Project Dashboard



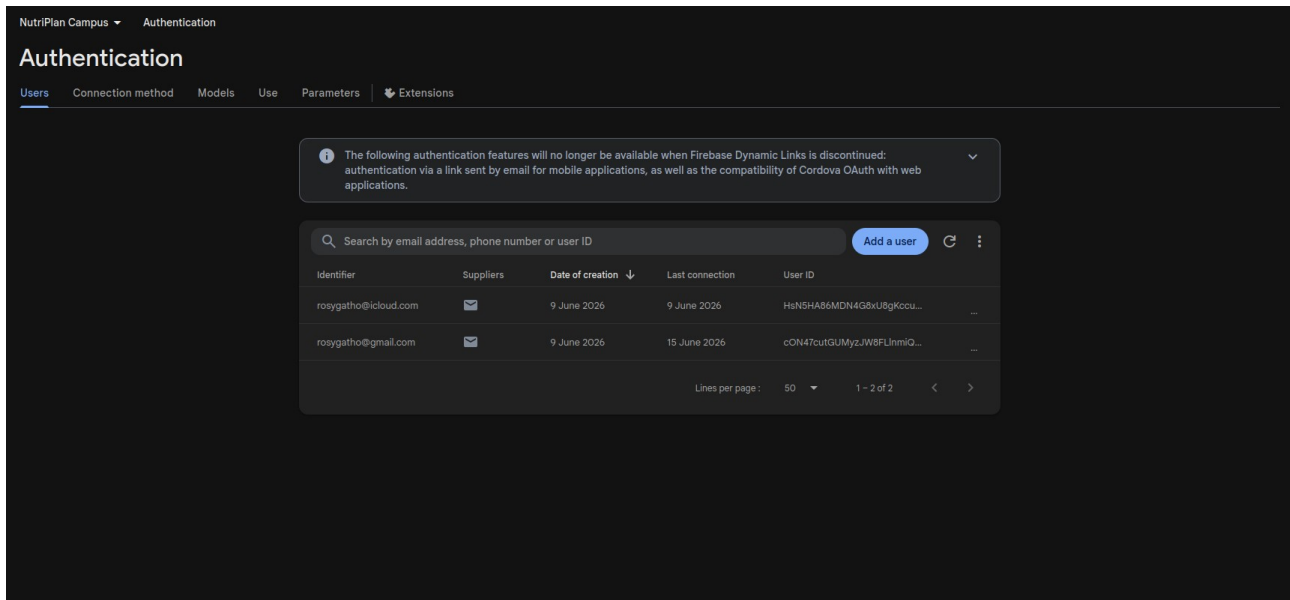
The NutriPlan application is connected to Firebase, a cloud-based backend platform provided by Google. Firebase provides services such as Authentication, Firestore Database, Analytics, and Hosting. These services enable communication between the mobile application and remote servers over the internet.

Figure 2: Firestore Usage Statistics



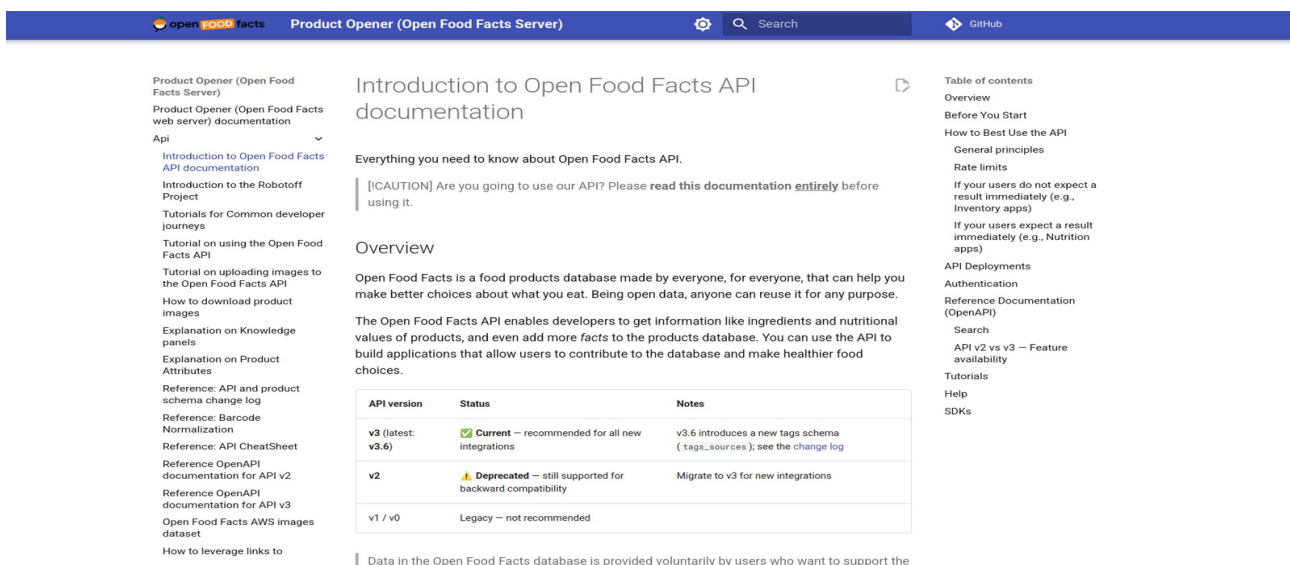
Cloud Firestore records the number of read and write operations performed by the NutriPlan application. During application usage, 115 read operations and 1 write operation were recorded. These operations demonstrate how the mobile application communicates with a cloud database through network requests and responses.

Figure 3: Firebase Authentication Users



Firebase Authentication is used to manage user registration and login in the NutriPlan application. User credentials are securely transmitted to Firebase servers for authentication. Registered users are stored in Firebase Authentication and can access the application using their credentials.

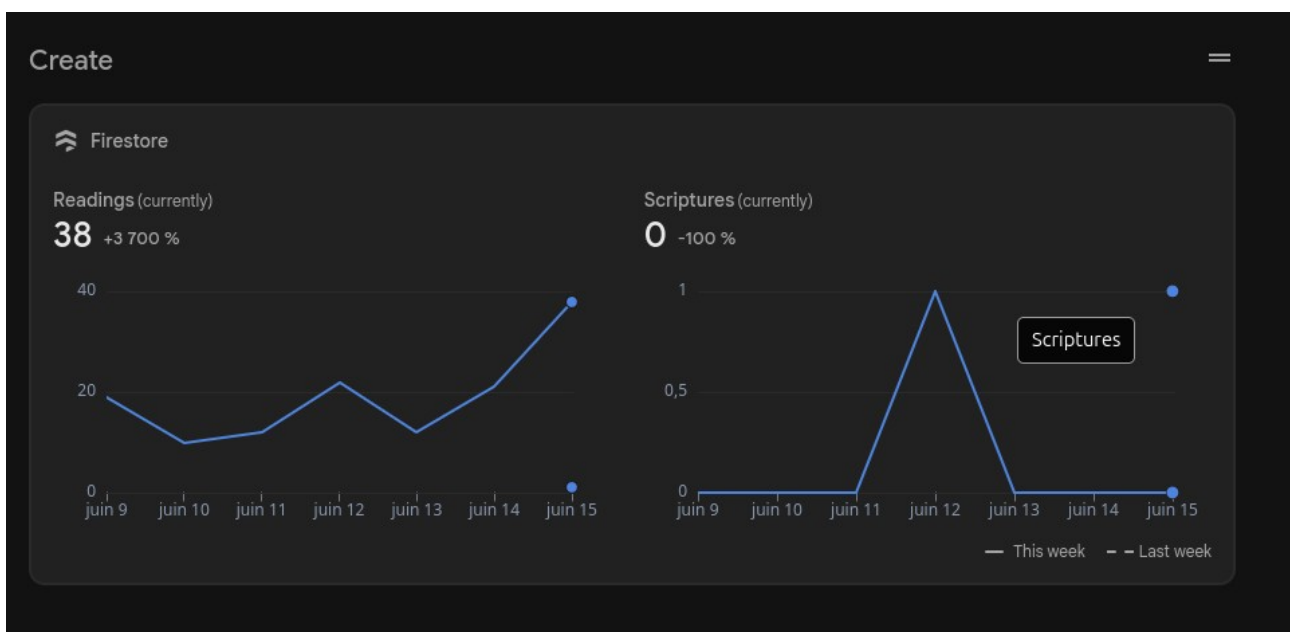
Figure 4: Open Food Facts API Documentation



The Open Food Facts API allows applications to retrieve food information, ingredients, and nutritional values from online servers. This enables mobile applications to access real-time food data through internet communication.

WEEK 7: DATA STORAGE AND DATABASE INTEGRATION IN MOBILE APPLICATIONS

Figure 5: Firestore Database Usage



Cloud Firestore is the database used by the NutriPlan application. The database records read and write operations performed by the application when users interact with the system.

Figure 6: Firestore Collection Structure

The Students collection stores application data in Firebase Firestore. Each document contains information entered by users and is uniquely identified by a document ID generated by Firebase.

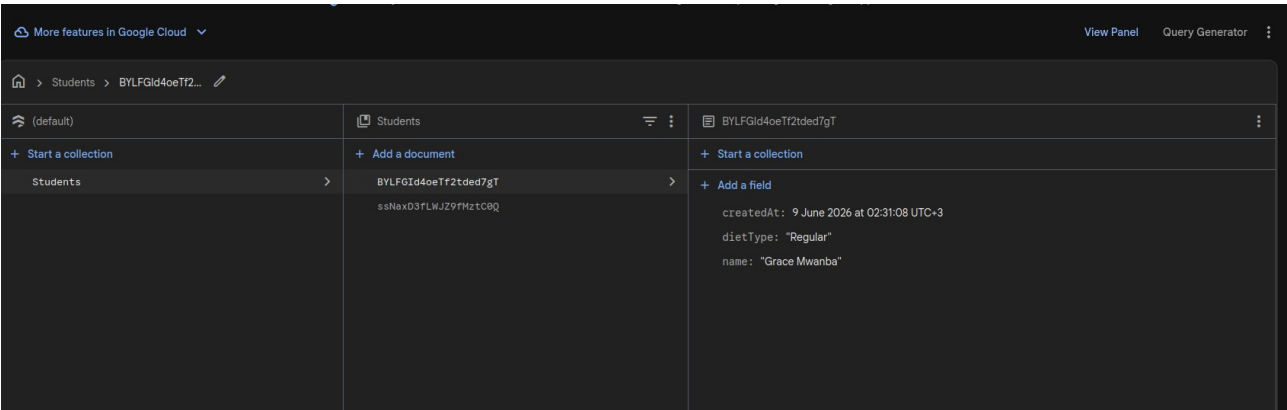
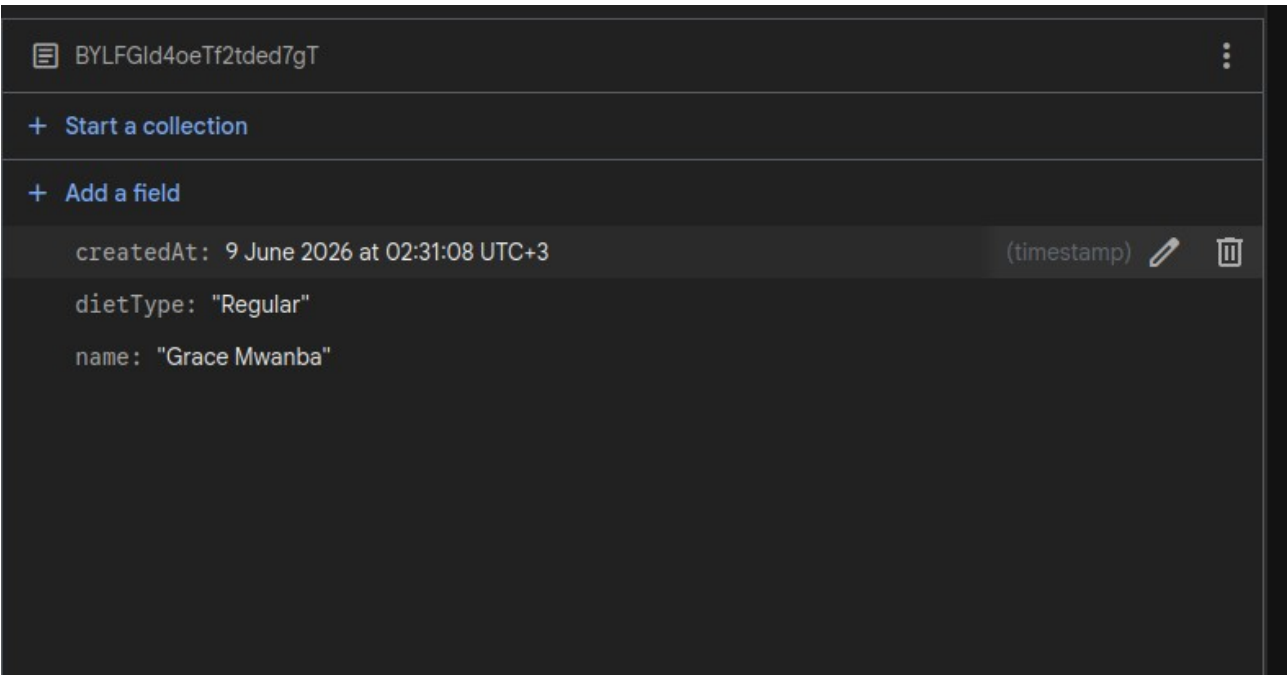
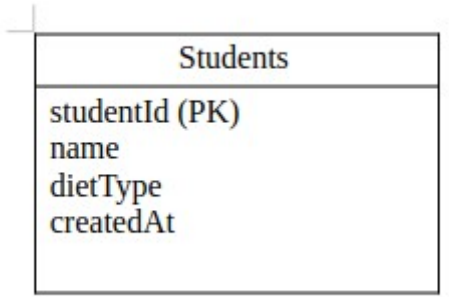


Figure 7: Firestore Document Data



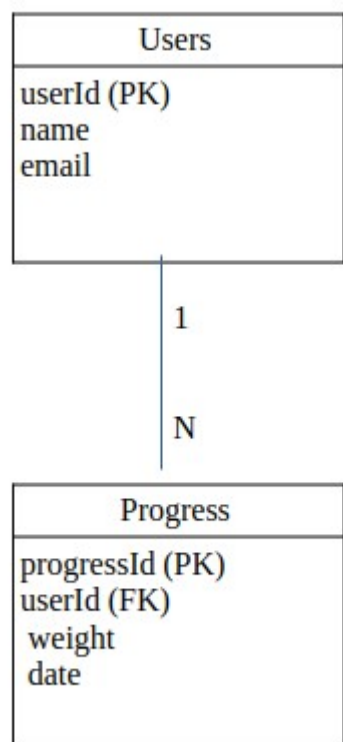
The document contains stored records in the database. Each record includes a user name, diet type, and creation date. This demonstrates successful storage and retrieval of information from the cloud database.

Figure 8: Students Collection Structure



The Students collection stores user information. The `studentId` acts as the primary key and uniquely identifies each record in the database.

Figure 9: Relationship Diagram



The relationship between Users and Progress is one-to-many (1:N). One user can have multiple progress records. The `userId` field in the Progress collection acts as a foreign key linking progress records to a specific user.